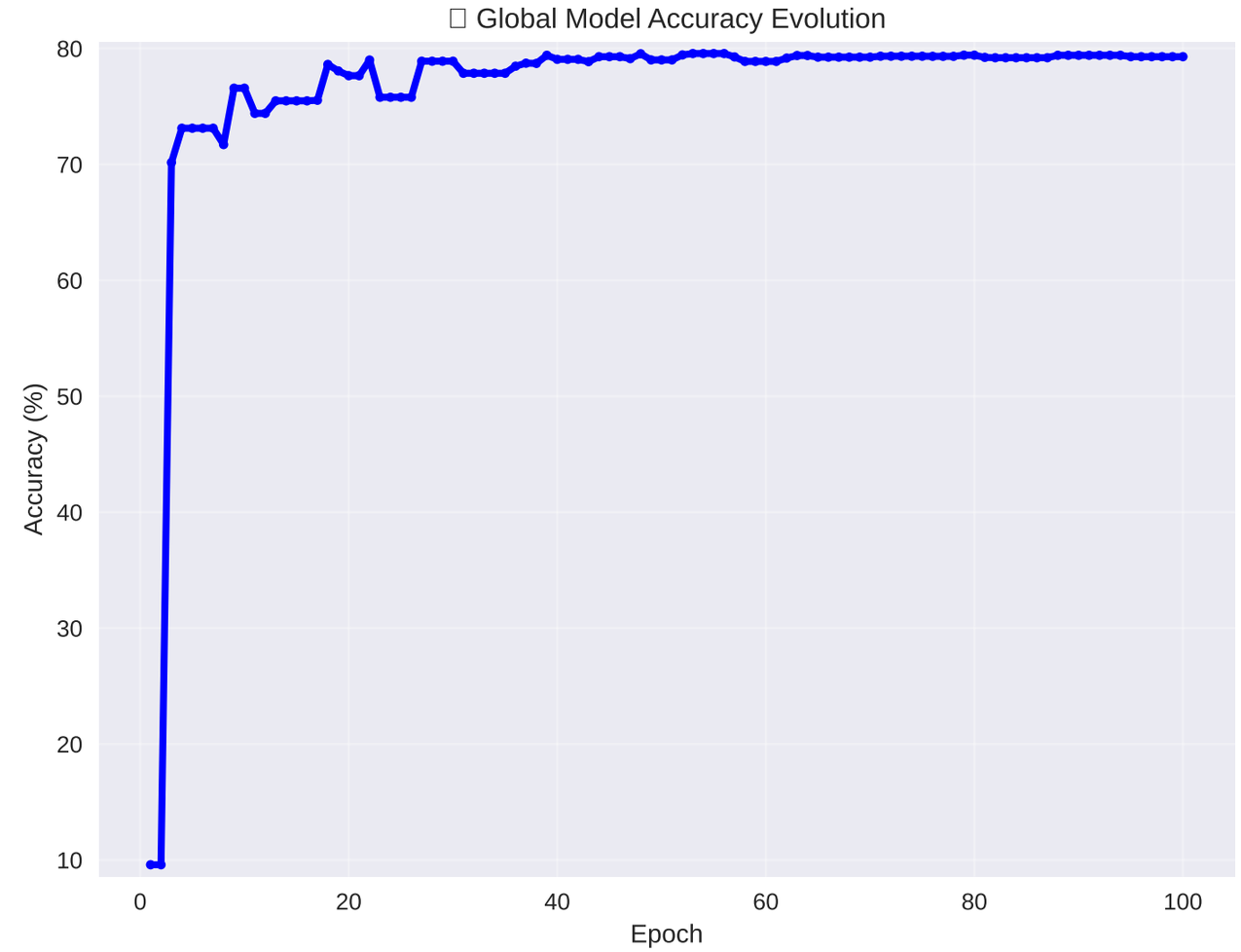
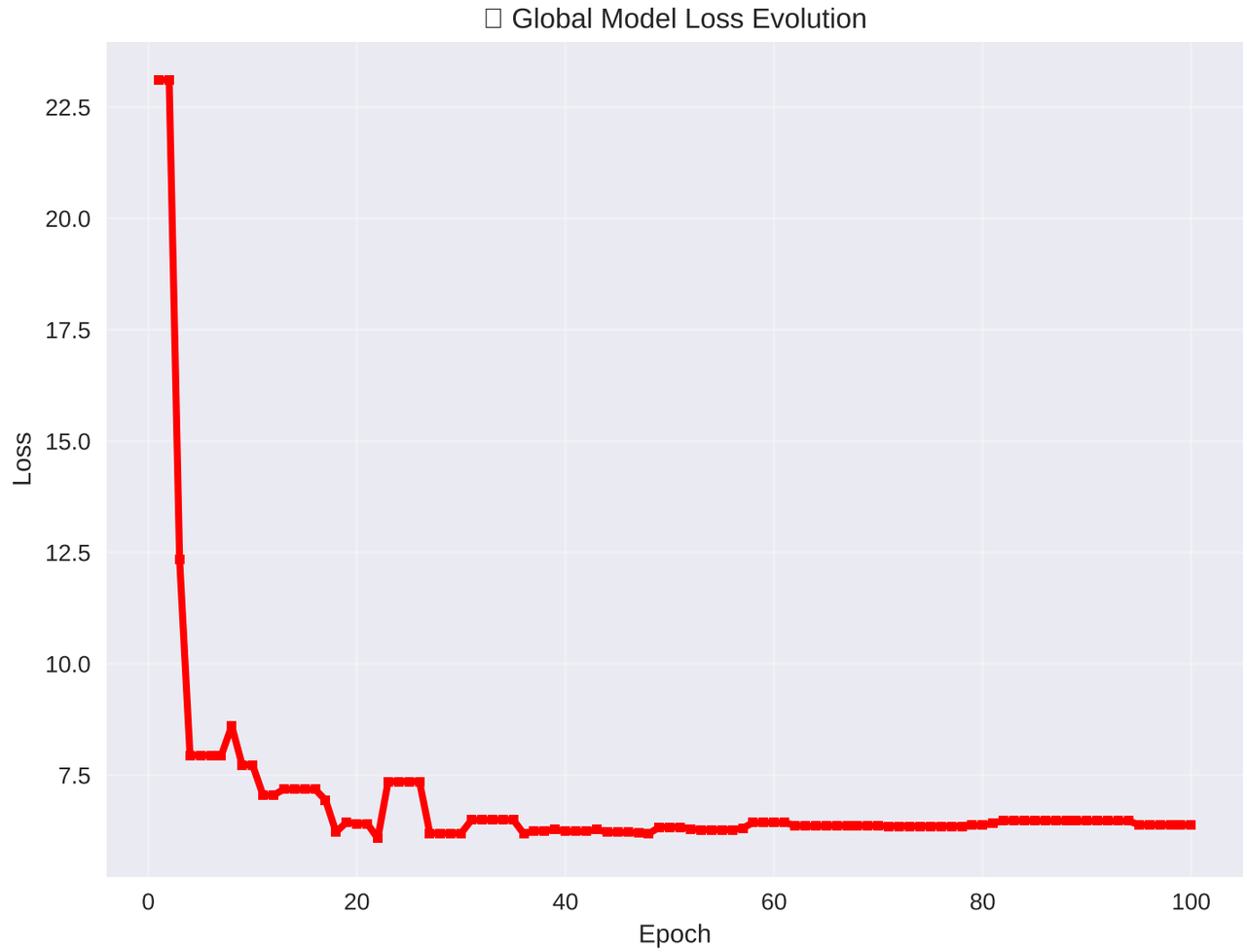
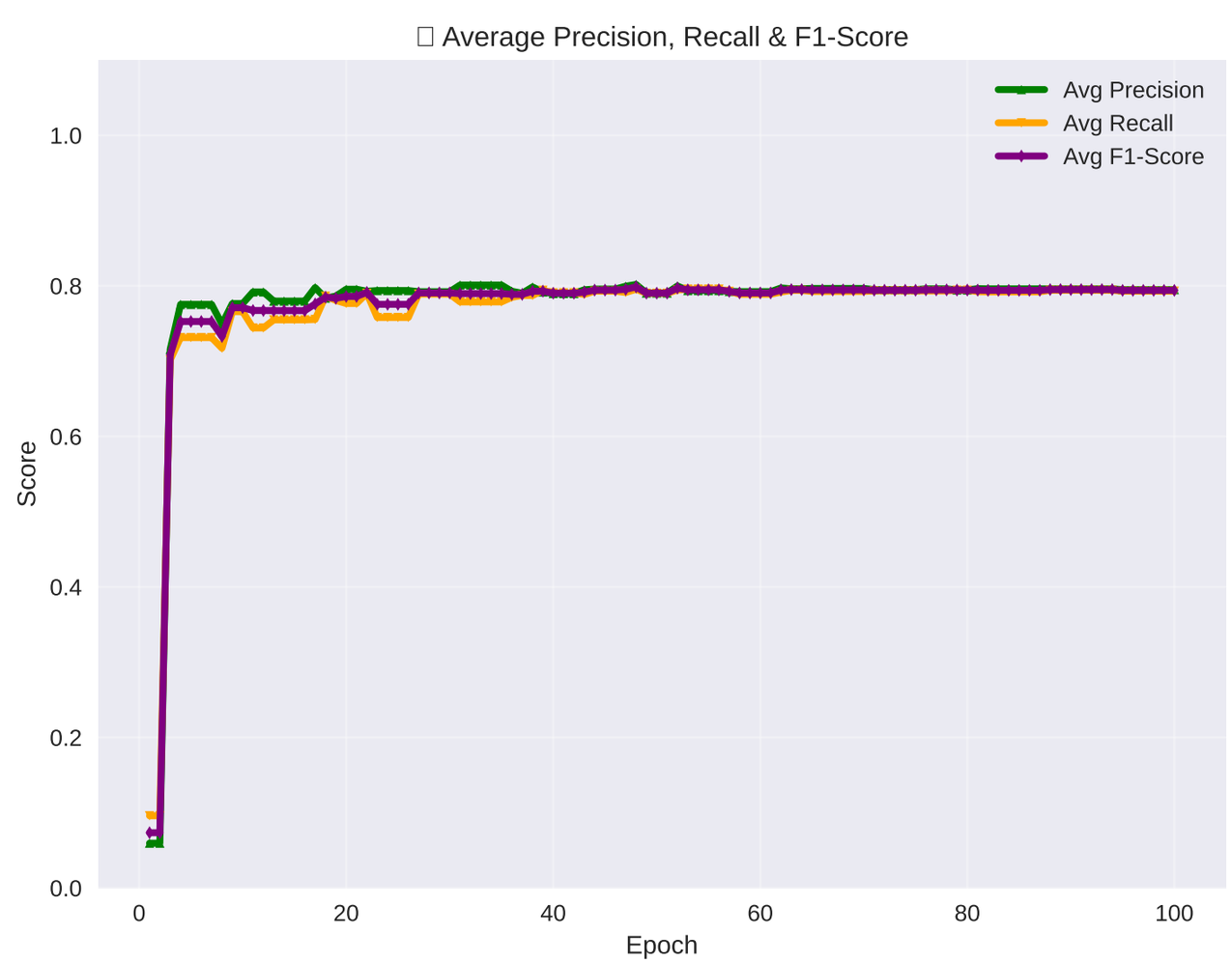
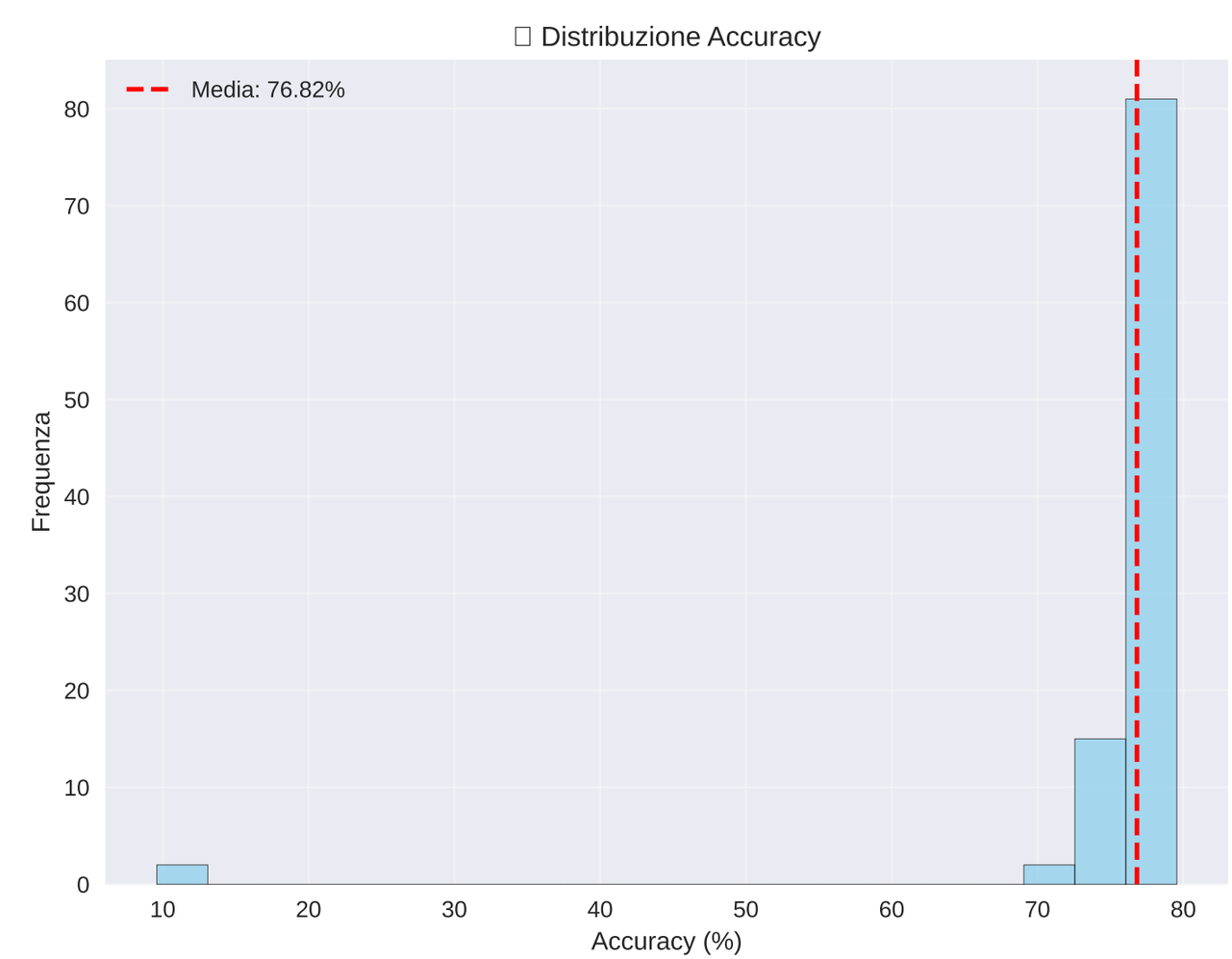
Federated Learning - Main Metrics Overview



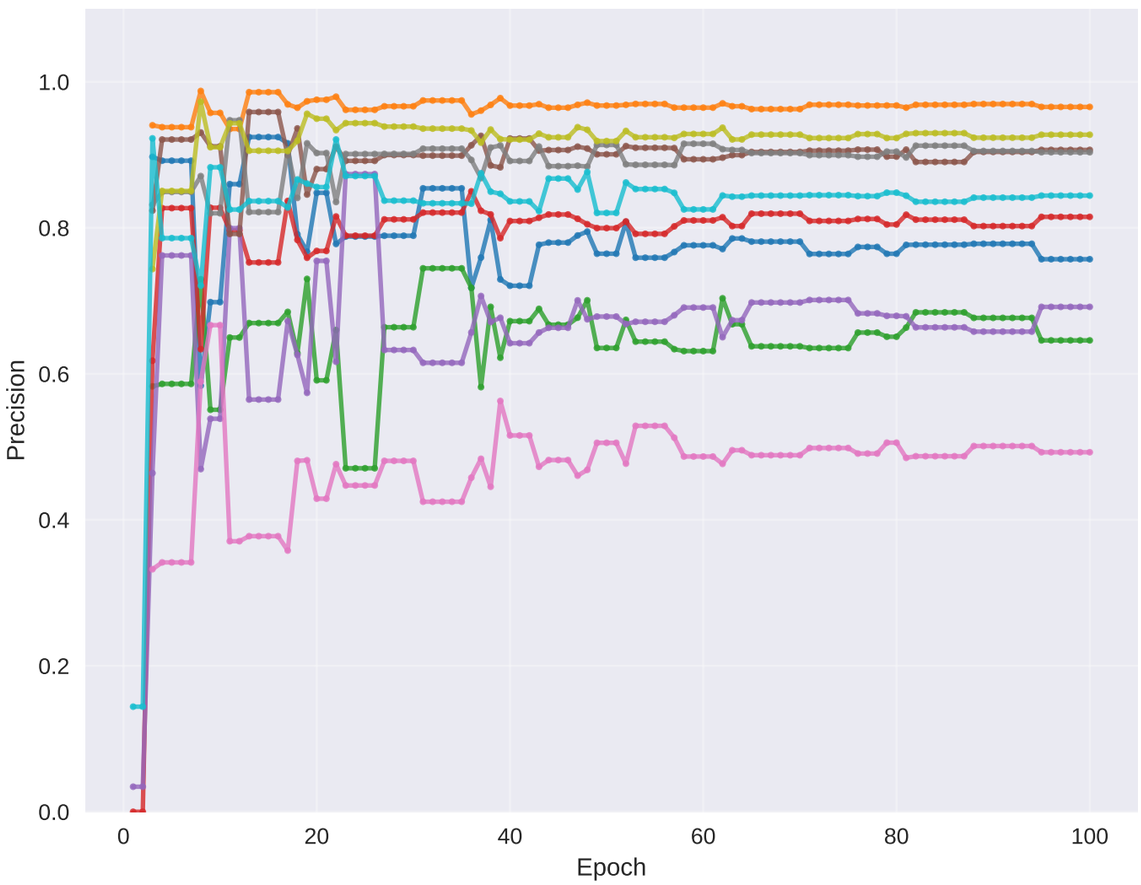




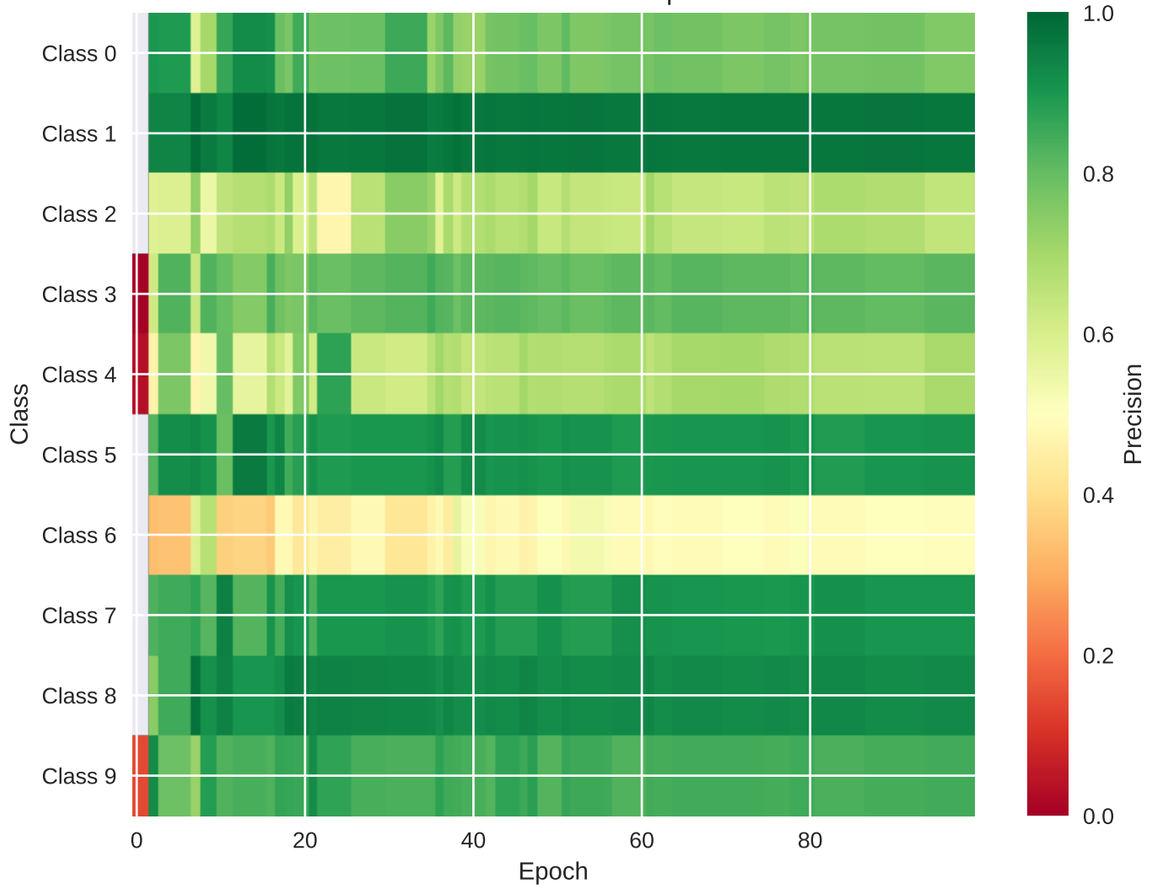


Per-Class Precision Analysis

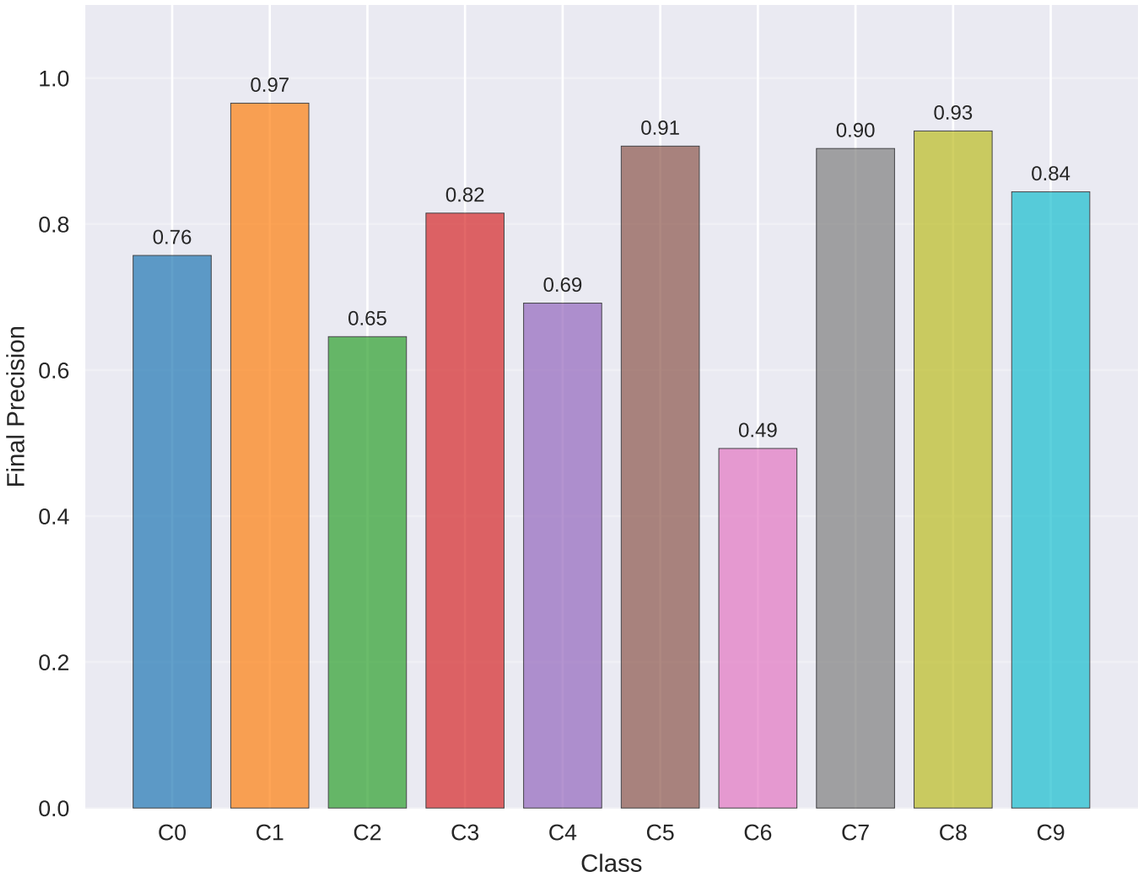
□ Per-Class Precision Evolution



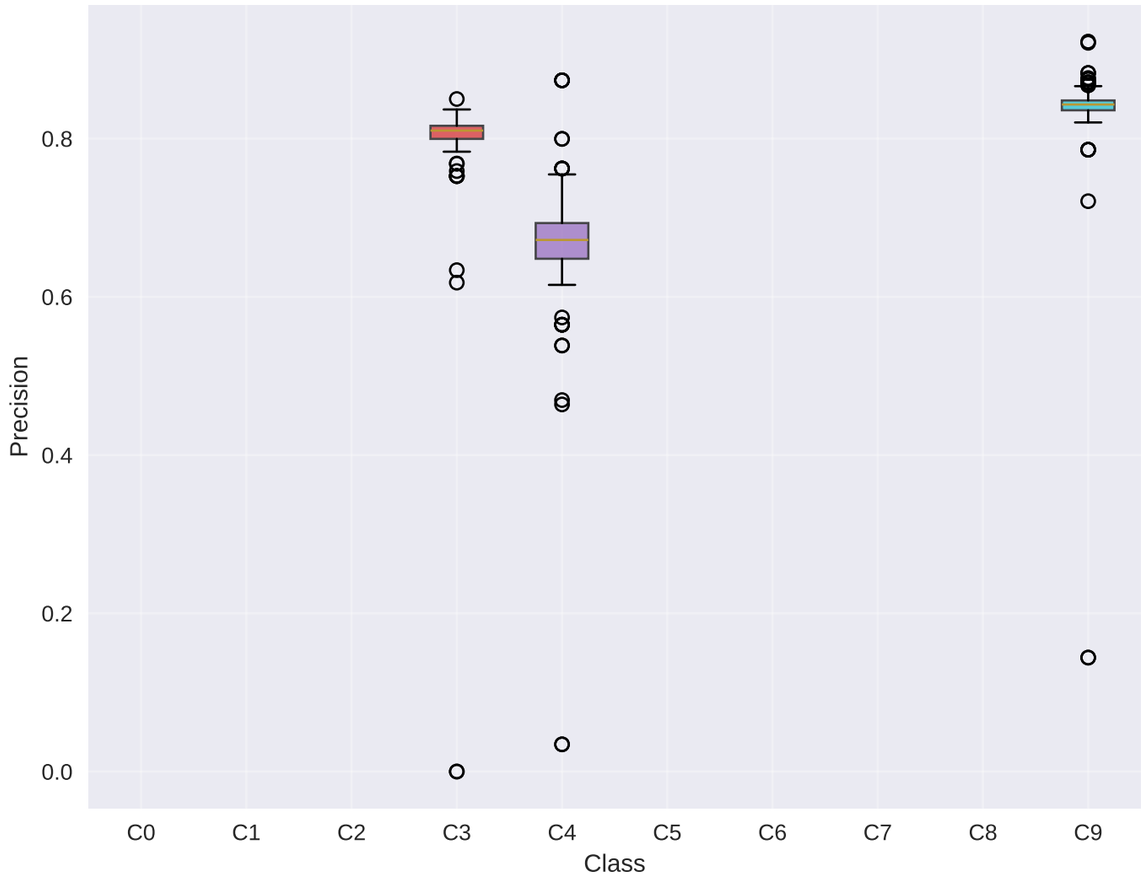
□ Precision Heatmap



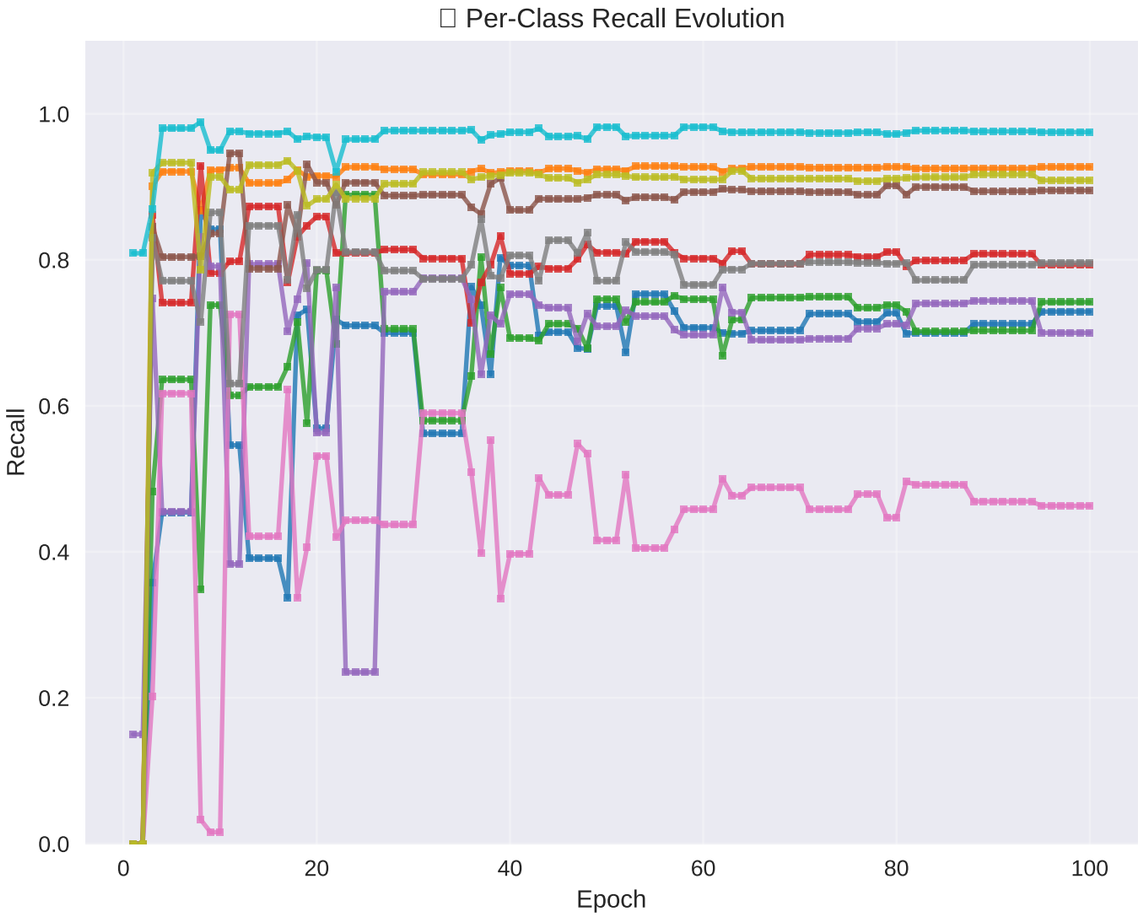
□ Final Precision per Class

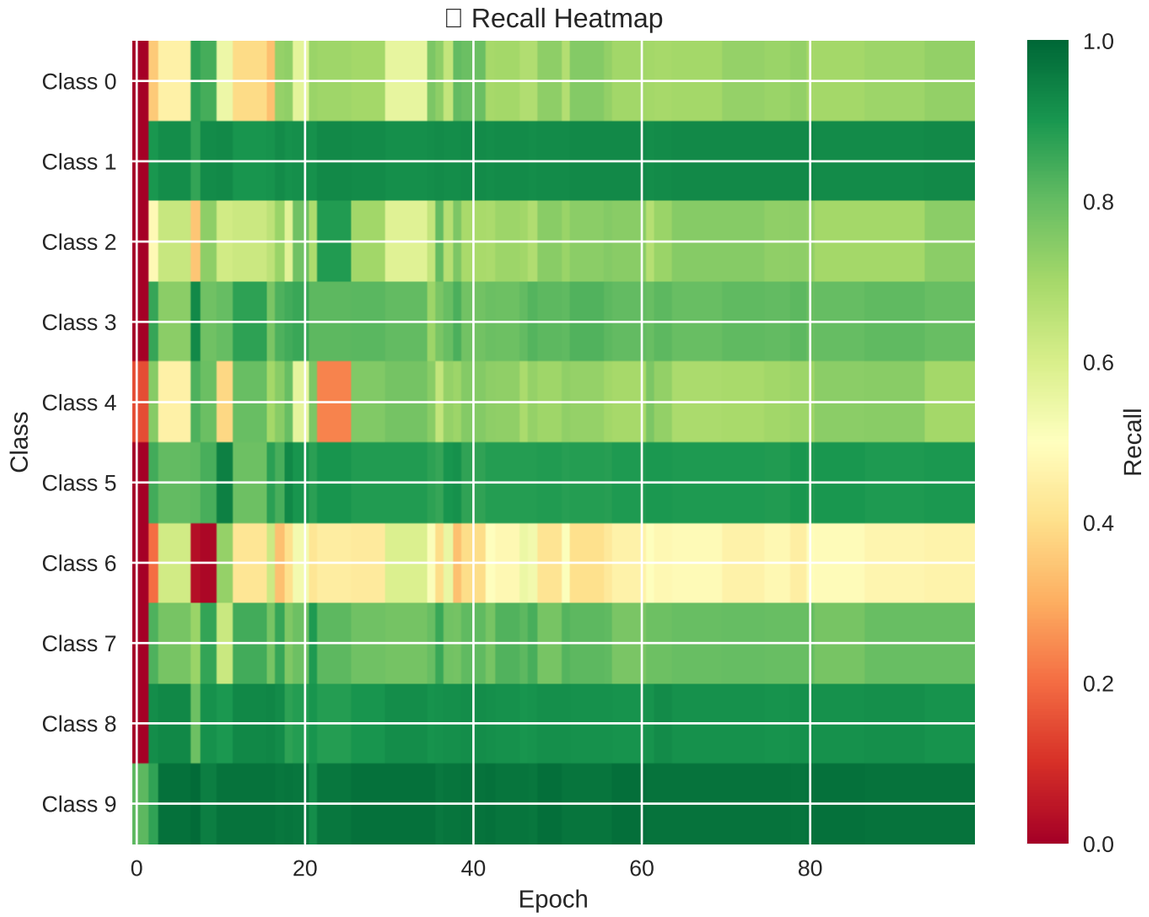


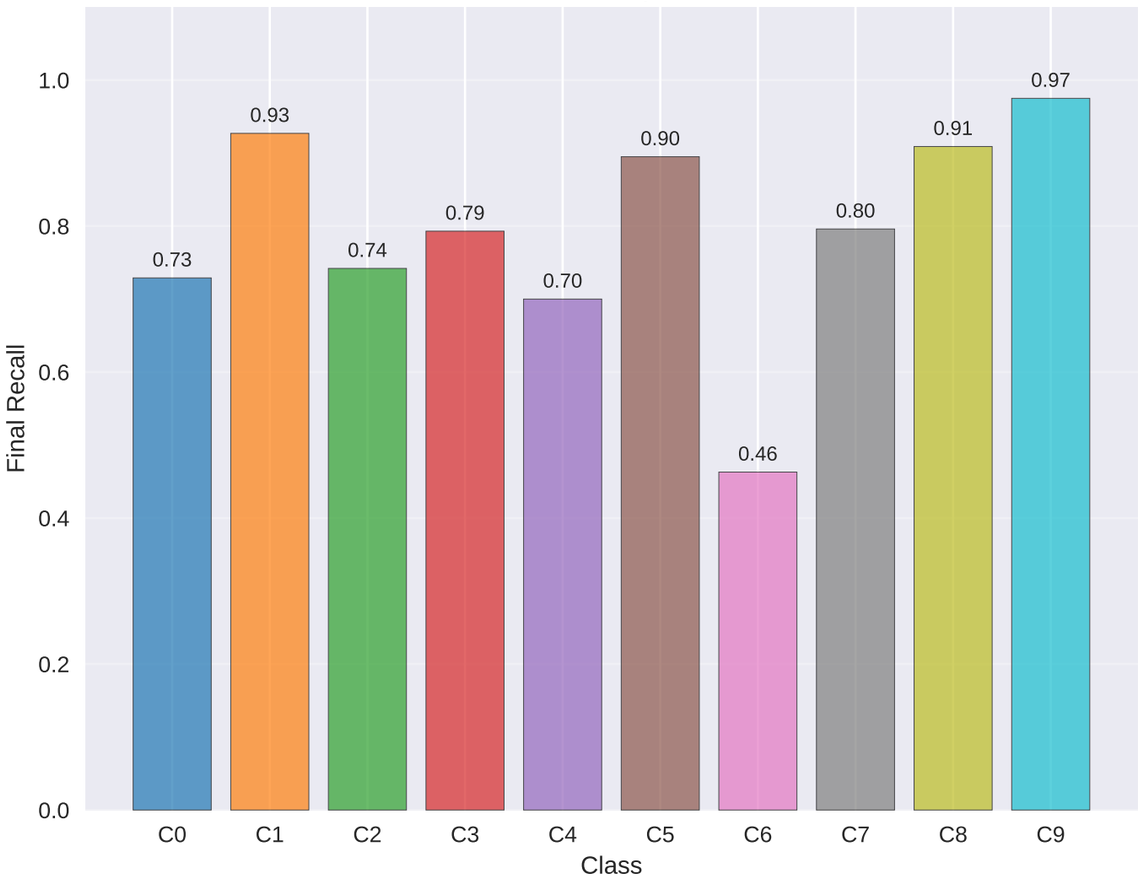
□ Precision Distribution per Class

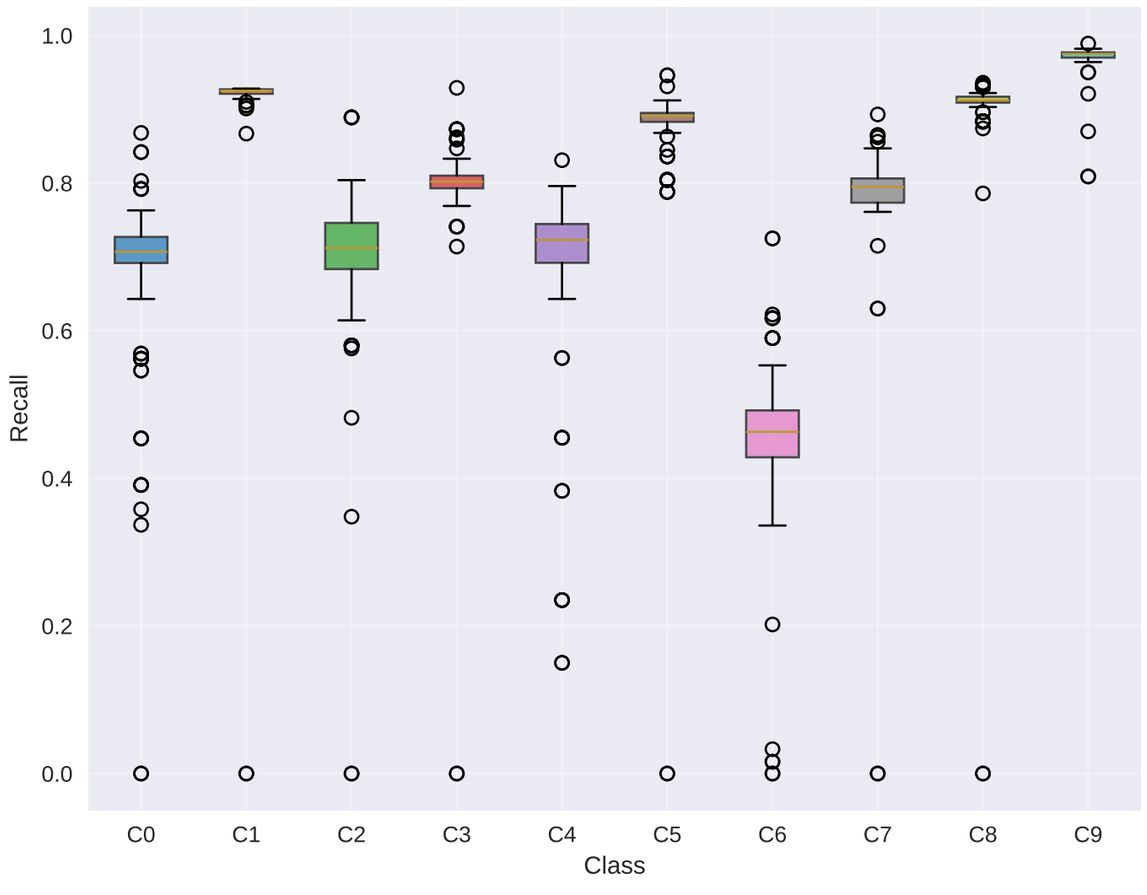


Per-Class Recall Analysis

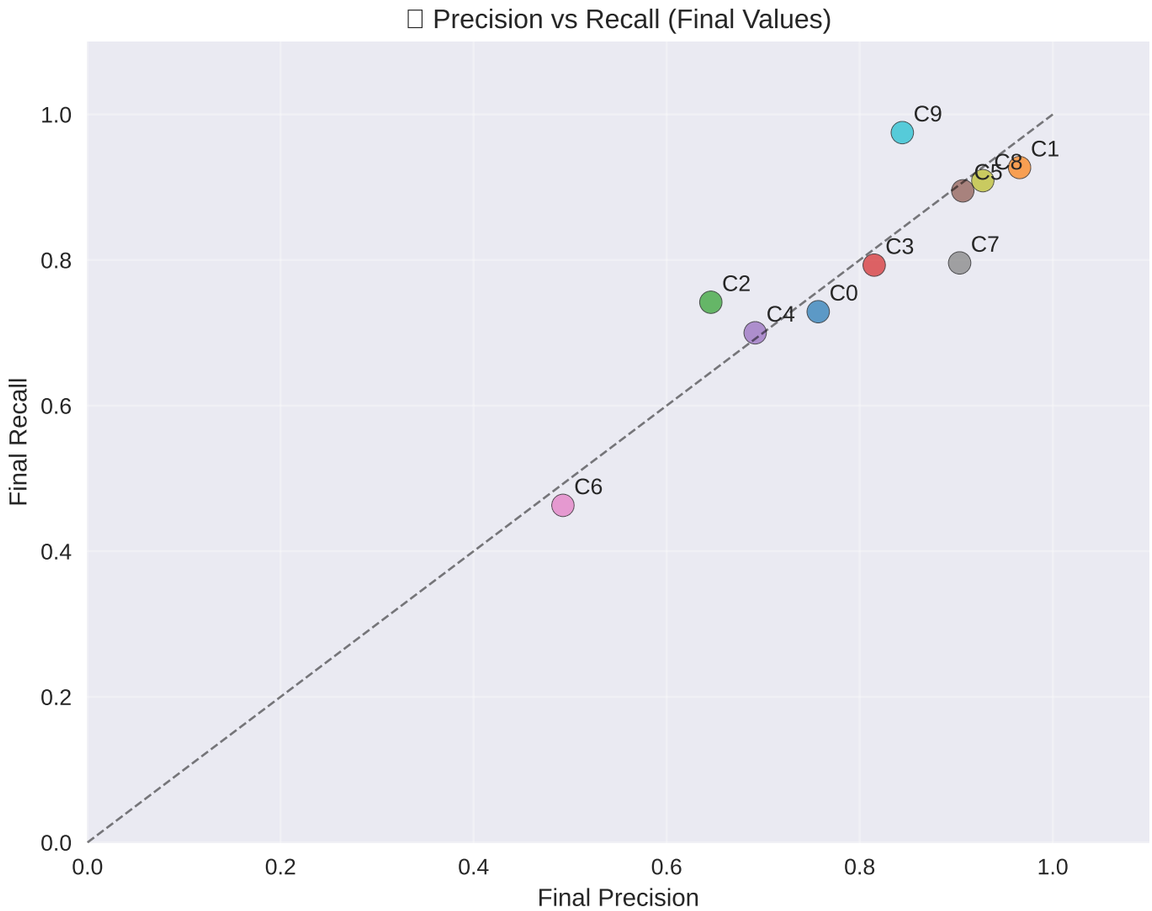


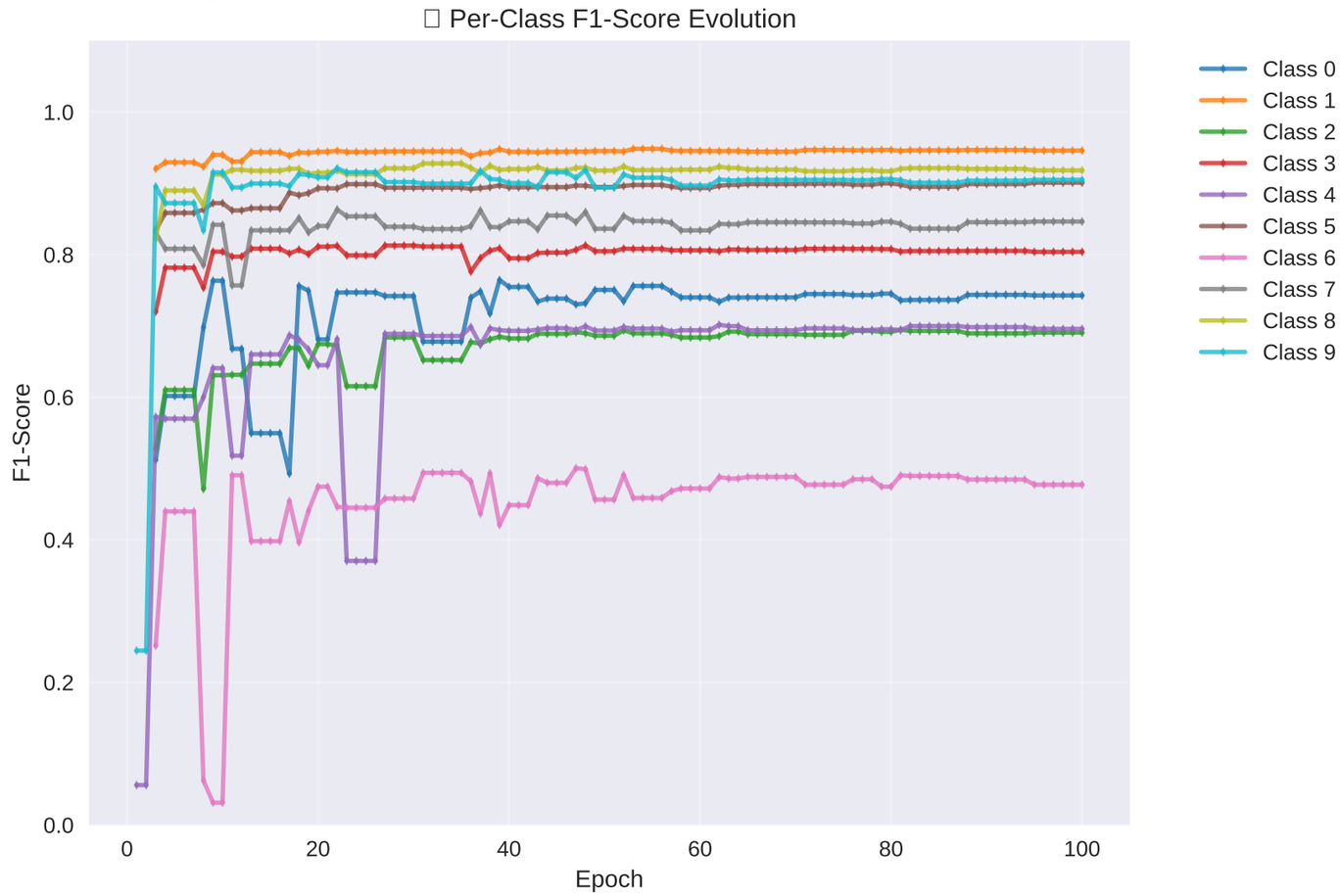


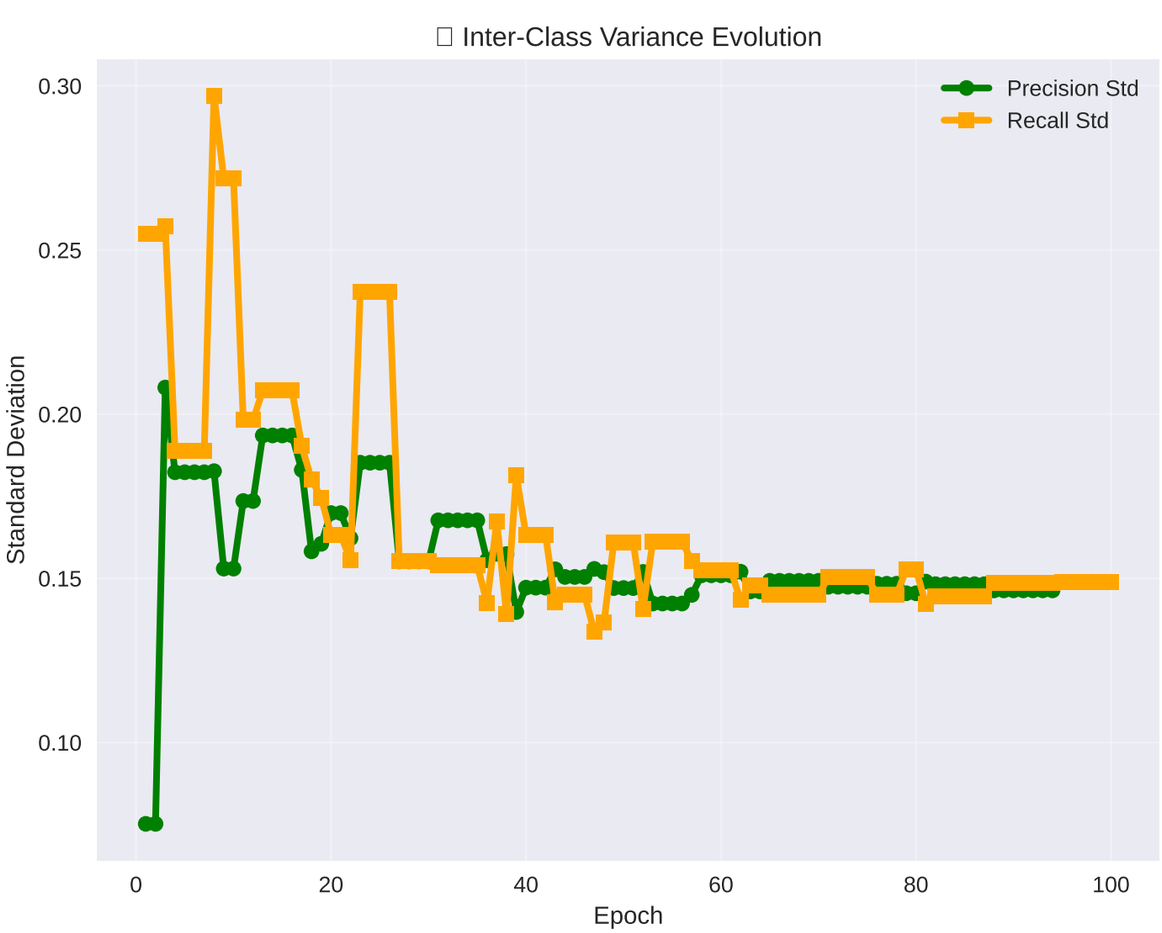


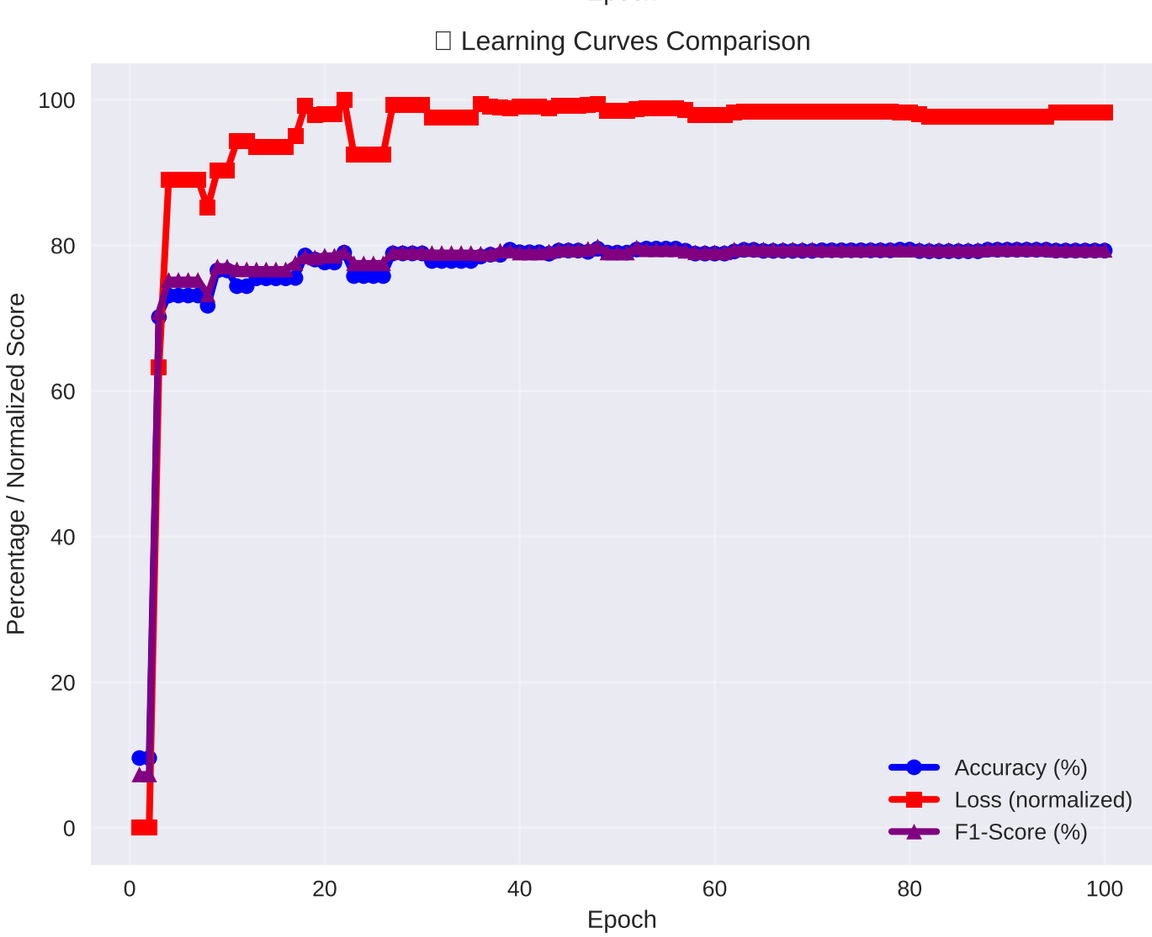


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-09-12 14:04:38
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 79.29%
• Loss: 6.3819
• Average Precision: 0.795
• Average Recall: 0.793
• Average F1-Score: 0.794

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +69.70%
• Riduzione Loss: +16.7285
• Stabilità Accuracy (CV): 0.1282

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.966)
• Peggiore Precision: Classe 6 (0.493)
• Migliore Recall: Classe 9 (0.975)
• Peggiore Recall: Classe 6 (0.463)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.757, R=0.729, F1=0.743
• Classe 1: P=0.966, R=0.927, F1=0.946
• Classe 2: P=0.646, R=0.742, F1=0.691
• Classe 3: P=0.815, R=0.793, F1=0.804
• Classe 4: P=0.692, R=0.700, F1=0.696
• Classe 5: P=0.907, R=0.895, F1=0.901
• Classe 6: P=0.493, R=0.463, F1=0.477
• Classe 7: P=0.904, R=0.796, F1=0.846
• Classe 8: P=0.928, R=0.909, F1=0.918
• Classe 9: P=0.844, R=0.975, F1=0.905

VALUTAZIONE QUALITATIVA:
• Performance Globale: DA MIGLIORARE
• Stabilità Training: INSTABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
• Considerare aumento epoche o tuning iperparametri
• Implementare class balancing o weighted loss
• Migliorare stabilità con learning rate scheduling
```